

# DSSAT User Guide for 2D model

V2.2 2025-06-10

## Resources

Source code: [https://github.com/chporter/dssat-csm-os/tree/2D\\_integrated](https://github.com/chporter/dssat-csm-os/tree/2D_integrated)

Sample data set: [https://github.com/DSSAT/dssat-csm-data/tree/2D/2D\\_samples](https://github.com/DSSAT/dssat-csm-data/tree/2D/2D_samples)

## Description

The DSSAT-CSM 2D model was created for use in raised-bed, drip irrigation systems. It will work in either flat or raised-bed systems, with or without plastic mulch cover, and with sub-daily drip irrigation schedules. The geometry of the bed is as shown below. The model simulates half of a row, assuming symmetry around the centerline of the row. The user specifies a bed height and width and the bed is “constructed” from the soil profile by calculating a “dig depth” and mixing the soil properties in the bed from the soil layers within this depth. A cross section of the half-row is shown in Figure 1.

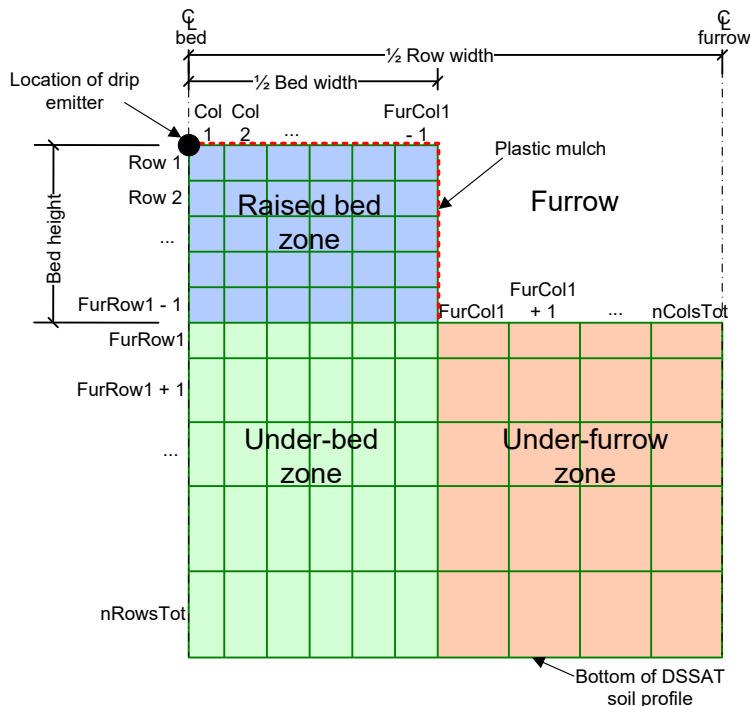


Figure 1. Cross section of a half-row showing the geometry of the bed and the rows and columns for modeling the 2D system.

The location of the drip irrigation emitters can be specified by an offset from the centerline and a depth from the surface. This places the irrigation water and fertigation N into a specific cell in the bed cross section. The drip emitters are assumed to be a line source, i.e., every point along a row is wetted evenly by the emitters.

## Input data

There are a few new inputs that are required to run the 2D model. These are:

- Bed height, bed width, plastic mulch albedo
- Drip irrigation schedule including location of drip emitter(s) and timing of irrigation.
- MEHYD switch in simulation controls = "G" to specify that a 2D simulation will be done.
- Van Genuchten soil parameters can be optionally entered into the soil profile data. These will be calculated by the model from standard DSSAT inputs if the user does not have values.
- Species parameters for CROPGRO crops can define the horizontal progression of root growth. If no parameters are included in the species file, then the vertical parameters are also used for horizontal growth.

The Sample 2D dataset which is available at the link under "Resources" at the top of this document, will be used as an example. The sample FileX contains a hypothetical "experiment" with five treatments describing the possible field and plastic mulch configurations that are handled by the model (below).

| *TREATMENTS |   |   |   |                     | -----FACTOR LEVELS----- |    |    |    |    |    |    |    |    |    |    |    |    |
|-------------|---|---|---|---------------------|-------------------------|----|----|----|----|----|----|----|----|----|----|----|----|
| @N          | R | O | C | TNAME.....          | CU                      | FL | SA | IC | MP | MI | MF | MR | MC | MT | ME | MH | SM |
| 1           | 1 | 1 | 0 | 2D, flat, No PM     | 1                       | 1  | 0  | 1  | 1  | 1  | 1  | 0  | 0  | 1  | 0  | 1  | 1  |
| 2           | 1 | 1 | 0 | 2D, flat, Part PM   | 1                       | 2  | 0  | 1  | 1  | 1  | 1  | 0  | 0  | 1  | 0  | 1  | 1  |
| 3           | 1 | 1 | 0 | 2D, flat, Full PM   | 1                       | 3  | 0  | 1  | 1  | 1  | 1  | 0  | 0  | 1  | 0  | 1  | 1  |
| 4           | 1 | 1 | 0 | 2D, raised, No PM   | 1                       | 4  | 0  | 1  | 1  | 1  | 1  | 0  | 0  | 1  | 0  | 1  | 1  |
| 5           | 1 | 1 | 0 | 2D, raised, Part PM | 1                       | 5  | 0  | 1  | 1  | 1  | 1  | 0  | 0  | 1  | 0  | 1  | 1  |

The "Fields" section of FileX shows how these bed configurations are specified (see below). Three variables describe the field configuration using a new, third tier in the fields section:

- PMALB - Albedo for plastic mulch (fraction)
- BDWD – width of raised bed or width of plastic mulch for flat systems (cm)
- BDHT – bed height (cm)

| *FIELDS |                 |          |      |                      |            |            |      |       |      |      |            |                     |
|---------|-----------------|----------|------|----------------------|------------|------------|------|-------|------|------|------------|---------------------|
| @L      | ID_FIELD        | WSTA.... | FLSA | FLOB                 | FLDT       | FLDD       | FLDS | FLST  | SLTX | SLDP | ID_SOIL    | FLNAME              |
| 1       | TS2D0001        | TS2D1501 | -99  | -99                  | -99        | -99        | -99  | -99   | SIL  | 160  | 2DTEST0001 | 2D, flat, No PM     |
| 2       | TS2D0001        | TS2D1501 | -99  | -99                  | -99        | -99        | -99  | -99   | SIL  | 160  | 2DTEST0001 | 2D, flat, Part PM   |
| 3       | TS2D0001        | TS2D1501 | -99  | -99                  | -99        | -99        | -99  | -99   | SIL  | 160  | 2DTEST0001 | 2D, flat, Full PM   |
| 4       | TS2D0001        | TS2D1501 | -99  | -99                  | -99        | -99        | -99  | -99   | SIL  | 160  | 2DTEST0001 | 2D, raised, No PM   |
| 5       | TS2D0001        | TS2D1501 | -99  | -99                  | -99        | -99        | -99  | -99   | SIL  | 160  | 2DTEST0001 | 2D, raised, Part PM |
| @L      | .....XCRD ..... |          |      |                      | YCRD ..... | ELEV ..... | AREA | SLLEN | FLWR | SLAS | FLHST      | FHDUR               |
| 1       | 40              |          |      |                      | 116        | 625        | -99  | -99   | -99  | -99  | -99        | -99                 |
| 2       | 40              |          |      |                      | 116        | 625        | -99  | -99   | -99  | -99  | -99        | -99                 |
| 3       | 40              |          |      |                      | 116        | 625        | -99  | -99   | -99  | -99  | -99        | -99                 |
| 4       | 40              |          |      |                      | 116        | 625        | -99  | -99   | -99  | -99  | -99        | -99                 |
| 5       | 40              |          |      |                      | 116        | 625        | -99  | -99   | -99  | -99  | -99        | -99                 |
| @L      | PMALB           | BDWD     | BDHT |                      |            |            |      |       |      |      |            |                     |
| 1       | -99             | -99      | -99  | !2D, flat, No PM     |            |            |      |       |      |      |            |                     |
| 2       | 0.3             | 35       | 0    | !2D, flat, Part PM   |            |            |      |       |      |      |            |                     |
| 3       | 0.3             | 70       | 0    | !2D, flat, Full PM   |            |            |      |       |      |      |            |                     |
| 4       | -99             | 35       | 25   | !2D, raised, No PM   |            |            |      |       |      |      |            |                     |
| 5       | 0.3             | 35       | 25   | !2D, raised, Part PM |            |            |      |       |      |      |            |                     |

For a 2D model, a raised bed configuration can be specified by combinations of bed width (BDWD) and bed height (BDHT). If either of these values is zero or -99, then a flat system (i.e., no raised bed) is

modeled. If a plastic mulch albedo is specified for a raised bed, it is assumed to cover the top and sides of the raised bed.

For a flat 2D simulation or a 1D simulation, the PMALB and BDWD variables can be used to describe a plastic mulch cover with an albedo of PMALB and a width of BDWD, where BDWD is less than or equal to the row spacing specified in the planting section of FileX.

Drip irrigation is specified by sub-daily irrigation records for one or more drip lines. Each drip line is described in a new, second tier of irrigation data. In the example below, there are two drip lines for irrigation level 1, one used for regular irrigation and one for fertigation. The location and spacing of the emitters are defined by the following variables:

- IRSPC – emitter spacing (cm)
- IROFS – emitter offset from centerline (cm)
- IRDEP – depth of emitter (cm)

```
*IRRIGATION AND WATER MANAGEMENT
@I EFIR IDEP ITHR IEPT IOFF IAME IAMT IRNAME
1 1 -99 -99 -99 -99 -99 -99
@I IRLN IRSPC IROFS IRDEP
1 1 40 0 0 !Irrigation line
1 2 30 15 10 !Fertigation line
@I IDATE IROP IRVAL IRSTR IRDUR IRLN
1 15124 IR005 0.383 8:00 434.6 1
1 15152 IR005 0.383 8:00 120 1
1 15152 IR005 0.383 17:00 120 1
1 15179 IR005 0.200 8:00 120 2
1 15179 IR005 0.383 17:00 240 1
1 15192 IR005 0.200 8:00 120 2
1 15192 IR005 0.383 17:00 240 1
1 15202 IR005 0.383 8:00 330.3 1
1 15218 IR005 0.200 8:00 120 2
1 15218 IR005 0.383 17:00 240 1
1 15231 IR005 0.200 8:00 120 2
1 15231 IR005 0.383 17:00 240 1
1 15240 IR005 0.383 8:00 173.8 1
1 15254 IR005 0.200 8:00 120 2
1 15254 IR005 0.383 17:00 240 1
```

The drip irrigation scheduling is specified in the 3<sup>rd</sup> tier of the irrigation section of FileX.

- IDATE – date of irrigation application (YRDOY)
- IROP - irrigation operation code. IR005 indicates drip irrigation.
- IRVAL – irrigation rate (ml/s for IR005, 2D drip irrig)
- IRSTR – irrigation start time in 24 hour clock format (HH:MM)
- IRDUR – irrigation duration (min)

Multiple irrigations per day may be entered using one or more drip lines.

Note that if a drip emitter is specified to be offset from the centerline, then the assumption of symmetry means that the model is considering that two drip lines are installed, equidistant from the row centerline.

In the Simulation Controls section of FileX, the HYDRO='G' option should be selected for the 2D simulation.

```
*SIMULATION CONTROLS
@N GENERAL      NYERS NREPS START SDATE RSEED SNAME..... SMODEL
1 GE            1      1      S 15121 2150 2D - HYDRO = 'G'
@N OPTIONS      WATER NITRO SYMBI PHOSP POTAS DISES  CHEM  TILL   CO2
1 OP            Y      Y      N      N      N      N      Y      M
@N METHODS      WTHER INCON LIGHT EVAPO INFIL PHOTO HYDRO NSWIT MESOM MESEV MESOL
1 ME            M      M      E      R      S      C      G      1      G      S      3
@N MANAGEMENT   PLANT IRRIG FERTI RESID HARVS
1 MA            R      R      R      R      R
@N OUTPUTS      FNAME OVVEW SUMRY FROPT GROUT CAOUT WAOUT NIOUT MIOUT DIOUT VBOSE CHOUT OPOUT
1 OU            N      Y      Y      1      Y      N      Y      Y      N      N      D      N      Y
```

In the soil file, a user may optionally supply van Genuchten parameters describing the water potential in the layered soil. CSM will compute these from water holding characteristics but better prediction will result if these parameters are known and supplied. The van Genuchten parameters are entered into a third tier of the soil profile data as shown below.

```
*2DTEST0001 SE      SL      160 Silty Loam
@SITE        COUNTRY    LAT      LONG SCS FAMILY
NM           China      40.      116. -99
@ SCOM SALB SLU1 SLDR SLRO SLNF SLPF SMHB SMPX SMKE
R .13 9 .10 61 0.95 1 B001 A001 B001
@ SLB SLMH SLLL SDUL SSAT SRGF SSKS SBDM SLOC SLCL SLSI SLCF SLNI SLHW SLHB SCEC SADC
20 -99 .183 .279 .356 1 1.59 1.41 1.71 27.8 35.4 -99 -99 8.5 -99 -99 -99
40 -99 .225 .348 .473 .449 .53 1.47 1.49 44.5 41.1 -99 -99 8.5 -99 -99 -99
60 -99 .206 .332 .454 .268 .68 1.53 1.27 18.1 62.5 -99 -99 8.5 -99 -99 -99
80 -99 .181 .325 .434 .147 .68 1.52 .84 21.7 70.5 -99 -99 8.5 -99 -99 -99
100 -99 .181 .325 .434 .105 .68 1.52 .42 22.8 68.1 -99 -99 8.5 -99 -99 -99
120 -99 .181 .325 .434 .101 .09 1.52 .11 22.8 68.1 -99 -99 8.5 -99 -99 -99
140 -99 .181 .325 .434 .014 .09 1.52 .1 22.8 68.1 -99 -99 8.5 -99 -99 -99
160 -99 .181 .325 .434 .014 .09 1.52 .1 22.8 68.1 -99 -99 8.5 -99 -99 -99
@ SLB SLPX SLPT SLPO CACO3 SLAL SLFE SLMN SLBS SLPA SLPB SLKE SLMG SLNA SLSU SLEC SLCA
20 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0
40 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0
60 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0
80 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0
100 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0
120 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0
140 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0
160 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0 -99.0
@ SLB ALFVG MVG NVG WCRES
20 0.011 0.159 1.189 0.070
40 0.016 0.145 1.170 0.050
60 0.015 0.143 1.167 0.023
80 0.010 0.166 1.199 0.020
100 0.010 0.166 1.199 0.020
120 0.010 0.166 1.199 0.020
140 0.010 0.166 1.199 0.020
160 0.010 0.166 1.199 0.020
```

Currently, the 2D model is not included in the release version of the DSSAT-CSM. The source code should be downloaded from the link provided above and compiled to create the 2D executable.

**WARNING! The new variables associated with the 2D model are NOT compatible with XBuild or SBuild. Please do not attempt to create your FileX with XBuild. If you open your 2D experimental file with XBuild, you may lose your data.**

For CROPGRO species, horizontal root growth parameters may be specified separately from the vertical root growth parameters. These parameters are defined as follows:

RTWIDI – Initial root width at emergence, cm

XRTFACH vs YRTFACH – Maximum rate of root horizontal progression, cm per physiological day

|                                |       |       |      |      |      |      |      |  |  |  |  |  |  |                                             |
|--------------------------------|-------|-------|------|------|------|------|------|--|--|--|--|--|--|---------------------------------------------|
| !*ROOT PARAMETERS              |       |       |      |      |      |      |      |  |  |  |  |  |  |                                             |
| 10.0                           | 9999. | 0.015 | 0.1  | 0.01 | 1.50 | 0.04 |      |  |  |  |  |  |  | RTDEPI,RFAC1,RTSEN,RLDSM,RTSDF,RWUEP1,RWUMX |
| 0.0                            | 2.85  | 3.0   | 2.85 | 6.0  | 2.85 | 30.0 | 2.85 |  |  |  |  |  |  | XRTFAC,YRTFAC                               |
| 0.012                          | 0.012 | 0.02  | 0.10 |      |      |      |      |  |  |  |  |  |  | RTNO3,RTNH4,PORMIN,RTEXF                    |
| !parameters added for 2D model |       |       |      |      |      |      |      |  |  |  |  |  |  |                                             |
| 5.0                            |       |       |      |      |      |      |      |  |  |  |  |  |  | RTWIDI                                      |
| 0.0                            | 0.71  | 3.0   | 0.71 | 6.0  | 0.71 | 30.0 | 0.71 |  |  |  |  |  |  | XRTFACH,YRTFACH                             |

If these parameters are missing from the species file for CROPGRO crops, the vertical root growth parameters are also used for horizontal root growth progression.

## Output data

The 2D model produces the same outputs as the standard 1D model, but additional detailed outputs can be generated by switching to the detailed verbosity output mode as shown in the box below.

|                      |       |       |       |       |       |                  |       |       |       |       |       |       |       |        |
|----------------------|-------|-------|-------|-------|-------|------------------|-------|-------|-------|-------|-------|-------|-------|--------|
| *SIMULATION CONTROLS |       |       |       |       |       |                  |       |       |       |       |       |       |       |        |
| @N GENERAL           | NYERS | NREPS | START | SDATE | RSEED | SNAME.....       |       |       |       |       |       |       |       | SMODEL |
| 1 GE                 | 1     | 1     | S     | 15121 | 2150  | 2D - HYDRO = 'G' |       |       |       |       |       |       |       |        |
| @N OPTIONS           | WATER | NITRO | SYMBI | PHOSP | POTAS | DISES            | CHEM  | TILL  | CO2   |       |       |       |       |        |
| 1 OP                 | Y     | Y     | N     | N     | N     | N                | N     | Y     | M     |       |       |       |       |        |
| @N METHODS           | WTHER | INCON | LIGHT | EVAP0 | INFIL | PHOTO            | HYDRO | NSWIT | MESOM | MESEV | MESOL |       |       |        |
| 1 ME                 | M     | M     | E     | R     | S     | C                | G     | 1     | G     | S     | 3     |       |       |        |
| @N MANAGEMENT        | PLANT | IRRIG | FERTI | RESID | HARVS |                  |       |       |       |       |       |       |       |        |
| 1 MA                 | R     | R     | R     | R     | R     |                  |       |       |       |       |       |       |       |        |
| @N OUTPUTS           | FNAME | OVVEW | SUMRY | FROPT | GROUT | CAOUT            | WAOUT | NIOUT | MIOUT | DIOUT | VBOSE | CHOUT | OPOUT |        |
| 1 OU                 | N     | Y     | Y     | 1     | Y     | N                | Y     | Y     | N     | N     | D     | N     | Y     |        |

The additional outputs are:

- SWcell\_60.csv – This file produces soil water content at 60-minute output intervals for every soil cell. (Note: for programmers: it is easy to modify the source code to produce output at smaller intervals.)
- SoilWat\_ts.OUT – Time step output and soil water balance.
- CellPlotDetail\_aaa.OUT – Daily output file for various soil water and soil N variables for each soil cell. This can be used to drive the 2D soil variable animated visualization tool. 'aaa' represents the treatment number up to 999 treatments.

These files can become very large (>1GB) for even a single treatment. It is advised that a user only run one treatment at a time when the detailed output is requested for a 2D simulation.